Covariant Form of Ohm's Law

Source: D. Tong, Electromagnetism, Problem Sheet 3, Question 10.

For a general 4-velocity, written as $u^\mu = \gamma(c, \mathbf{v})$, show that

$$F_{\mu\nu}u^\nu = \gamma \begin{pmatrix} -\mathbf{E}\cdot\mathbf{v}/c \\ \mathbf{E}+\mathbf{v}\times\mathbf{B} \end{pmatrix}. \tag{1.1}$$

In the rest-frame of a conducting medium, Ohm's law states that $\mathbf{J} = \sigma\mathbf{E}$, where σ is the conductivity and $\mathbf{J}$ is the 3-current. Assuming that σ is a Lorentz scalar, show that Ohm's law can be written covariantly as

$$J^\mu + \frac{1}{c^2}(J^\nu u_\nu)u^\mu = \sigma F^{\mu\nu}u_\nu, \tag{1.2}$$

where J^μ is the 4-current and u^μ is the (uniform) 4-velocity of the medium. If the medium moves with 3-velocity $\mathbf{v}$ in some inertial frame, show that the current in that frame is

$$\mathbf{J} = \rho\mathbf{v} + \sigma\gamma\left(\mathbf{E}+\mathbf{v}\times\mathbf{B} - \frac{1}{c^2}(\mathbf{v}\cdot\mathbf{E})\mathbf{v}\right), \tag{1.3}$$

where ρ is the charge density. Simplify this formula, given that the charge density vanishes in the rest-frame of the medium.

Solution. It is easy to verify that

$$F_{\mu\nu}u^\nu = \gamma \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix} \begin{pmatrix} c \\ v_x \\ v_y \\ v_z \end{pmatrix} = \gamma \begin{pmatrix} -\mathbf{E}\cdot\mathbf{v}/c \\ \mathbf{E}+\mathbf{v}\times\mathbf{B} \end{pmatrix}. \tag{1.4}$$

In the rest-frame, $u^\mu = (c, 0, 0, 0)$. Then, consider

$$J^\mu + \frac{1}{c^2}(J^\nu u_\nu)u^\mu = \begin{pmatrix} \rho c \\ \mathbf{J} \end{pmatrix} + \frac{1}{c^2}(-\rho c \quad \mathbf{J}) \cdot \begin{pmatrix} c \\ 0 \end{pmatrix} \begin{pmatrix} c \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \mathbf{J} \end{pmatrix}, \tag{1.5}$$

and

$$\sigma F_{\mu\nu}u^\nu = \begin{pmatrix} 0 \\ \sigma\mathbf{E}. \end{pmatrix} \tag{1.6}$$

Thus, we have verified that

$$J^\mu + \frac{1}{c^2}(J^\nu u_\nu)u^\mu = \sigma F^{\mu\nu}u_\nu \tag{1.7}$$

is indeed the covariant representation of Ohm's law.

For a moving medium, the equation gives

$$\begin{pmatrix} \rho c + \gamma^2(-\rho c^2 + \mathbf{J} \cdot \mathbf{v}) \\ \mathbf{J} + \gamma^2(-\rho c^2 + \mathbf{J} \cdot \mathbf{v})\mathbf{v}/c \end{pmatrix} = \gamma\sigma \begin{pmatrix} \mathbf{E} \cdot \mathbf{v}/c \\ \mathbf{E} + \mathbf{v} \times \mathbf{B} \end{pmatrix}. \tag{1.8}$$

Combining the two equations, we find

$$\mathbf{J} = \rho\mathbf{v} + \sigma\gamma\left(\mathbf{E} + \mathbf{v} \times \mathbf{B} - \frac{1}{c^2}(\mathbf{v} \cdot \mathbf{E})\mathbf{v}\right). \tag{1.9}$$

Finally, given $\rho_0 = 0$, then $J^\mu u_\mu = -\rho_0 c^2$. This implies

$$-\rho c^2 + \mathbf{J} \cdot \mathbf{v} = 0, \tag{1.10}$$

Taking the dot product of $\mathbf{J}$ with $\mathbf{v}$ gives

$$\rho v^2 + \sigma\gamma\left(\mathbf{v} \cdot \mathbf{E} - \frac{v^2}{c^2}\mathbf{v} \cdot \mathbf{E}\right) = \rho c^2. \tag{1.11}$$

Rearranging, we find

$$\rho = \frac{\sigma\gamma}{c^2}\,\mathbf{v} \cdot \mathbf{E}. \tag{1.12}$$

Substituting back, the current becomes

$$\mathbf{J} = \sigma\gamma\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{1.13}$$